

Question 1 (10 marks) Use a SEPARATE writing booklet **Marks**

- (a) Use $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of $y = x^2 - 3x$. **2**
- (b) Differentiate:
- (i) $3x^3 - \frac{x}{2} + 7$ **1**
- (ii) $\sqrt{x^2 - 1}$ **2**
- (iii) $\frac{x^2}{1 - 3x}$. **2**
- (c) Find the equation of the tangent to the curve $y = \frac{1}{x}$ at the point where $x = 2$. **3**

Question 2 (10marks) Use a SEPARATE writing booklet

- (a) If α and β are the roots of the equation $9x^2 + 3x - 2 = 0$, find the values of:
- (i) $\alpha + \beta$ **1**
- (ii) $\alpha\beta$ **1**
- (iii) $\alpha^2 + \beta^2$ **2**
- (b) Find the values of p for which the quadratic equation $x^2 + 3x + p = 0$ has equal roots. **2**
- (c) Solve for x $(2^x)^2 - 9(2^x) + 8 = 0$. **2**
- (d) Solve the inequality $x^2 - 4x > 0$. **2**

Question 3 (10marks) Use a SEPARATE writing booklet **Marks**

The equation of a curve is given by $y = x^3 - 3x^2 + 1$.

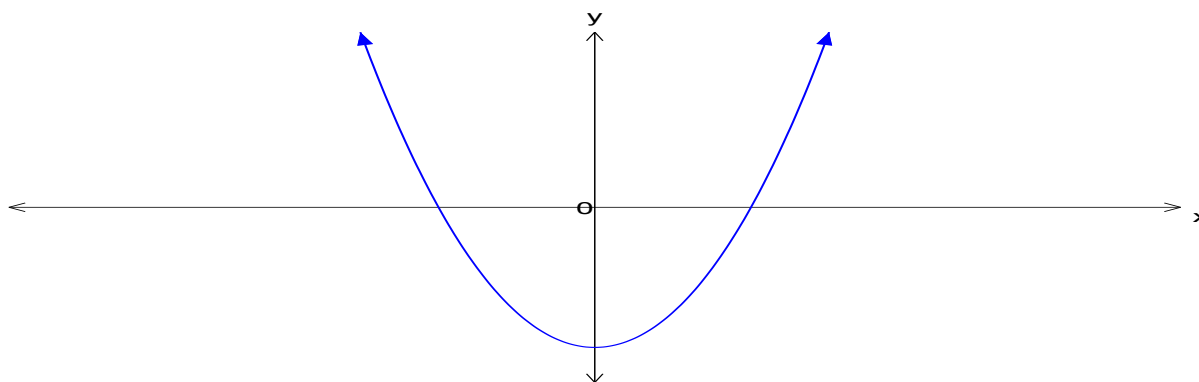
- (i) Find the coordinates of the stationary points and determine their nature. **3**
- (ii) Determine if there are any points of inflexion and if so, find their coordinates. **2**
- (iii) Sketch the curve for $-1 \leq x \leq 3$, showing stationary points and any points of inflexion. **2**
- (iv) For what values of x is the curve concave down? **1**
- (v) Use your sketch to determine what values of k will the equation $x^3 - 3x^2 + 1 - k = 0$ have 3 distinct solutions in the domain $-1 \leq x \leq 3$? **2**

Question 4 (10 marks) Use a **SEPARATE** writing booklet**Marks**

- (a) Determine the values of
- A
- ,
- B
- and
- C
- in

$$x^2 - 3x + 5 \equiv A(x-1)^2 + B(x-1) + C \quad 3$$

- (b) Copy the sketch of
- $y = f(x)$
- . Sketch a possible graph of
- $y = f'(x)$
- on it.
- 2

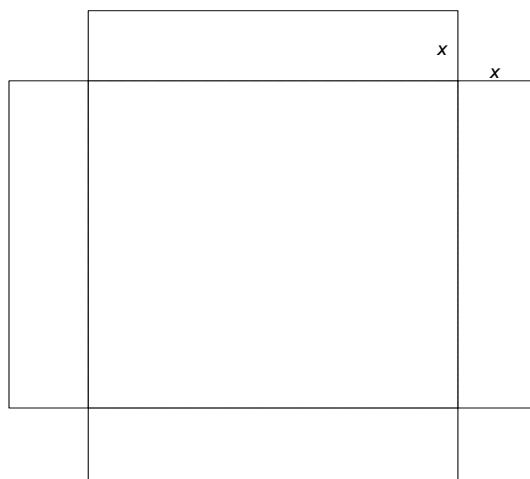


- (c) A
- 6 m
- by
- 6 m
- square sheet of metal has smaller
- $x\text{ m}$
- by
- $x\text{ m}$
- squares cut from its corners as shown below. The sheet is bent into an open box.

- (i) Use a diagram to help show that the volume of the box is given

$$\text{by } V(x) = x(6-2x)^2 \text{ m}^3. \quad 1$$

- (ii) What sized squares should be cut out to produce the box of greatest capacity?
- 4



Question 1

Yr 11 Maths HSC Ass1 2010

(a) $y = x^2 - 3x$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) - x^2 + 3x}{h}$$
$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h - x^2 + 3x}{h}$$
$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 3)}{h}$$
$$= 2x - 3$$

(b) (i) $\frac{d}{dx} [3x^3 - \frac{x}{2} + 7] = 9x^2 - \frac{1}{2}$ ✓

(ii) $\frac{d}{dx} [(x^2 - 1)^{\frac{1}{2}}] = \frac{1}{2}(x^2 - 1)^{-\frac{1}{2}} \cdot 2x$ ✓

$$= \frac{x}{\sqrt{x^2 - 1}}$$
 ✓

(iii) $\frac{d}{dx} [\frac{x^2}{1-3x}] = \frac{(1-3x) \cdot 2x - x^2 \cdot (-3)}{(1-3x)^2}$ ✓

$$= \frac{2x - 6x^2 + 3x^2}{(1-3x)^2}$$
$$= \frac{2x - 3x^2}{(1-3x)^2}$$
 ✓

(c) $y = \frac{1}{x}$

$$y = x^{-1}$$
$$\frac{dy}{dx} = -x^{-2}$$
$$= -\frac{1}{x^2}$$
 ✓

$\therefore$ when $x = 2$, $\frac{dy}{dx} = -\frac{1}{4}$

$\therefore$ Equation of tangent

$$y - \frac{1}{2} = -\frac{1}{4}(x - 2)$$
 ✓

$$4y - 2 = -x + 2$$

$$x + 4y - 4 = 0$$
 ✓

Q2

(a) $9x^2 + 3x - 2 = 0$

(i) $\alpha + \beta = -\frac{1}{3}$ ✓

(ii) $\alpha\beta = -\frac{2}{9}$ ✓

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ ✓
 $= \frac{1}{9} + \frac{4}{9}$
 $= \frac{5}{9}$ ✓

(b) $x^2 + 3x + p = 0$
 $\Delta = 0$ equal roots
 $\therefore b^2 - 4ac = 0$ ✓
 $9 - 4p = 0$
 $p = \frac{9}{4}$ ✓

(c) $(2^x)^2 - 9(2^x) + 8 = 0$
 $(2^x - 8)(2^x - 1) = 0$
 $2^x = 8 \text{ or } 2^x = 1$
 $x = 0, 3$ ✓✓

(d) $x^2 - 4x > 0$
 $x(x - 4) > 0$
 $\therefore x > 4 \text{ or } x < 0$
✓ ✓

Q3 $y = x^3 - 3x^2 + 1$

(i) $\frac{dy}{dx} = 3x^2 - 6x$

let $\frac{dy}{dx} = 0$ to find st pts

$$3x(x-2) = 0$$

$$x = 0, 2$$

$\therefore$ st pts at $(0, 1)$ and $(2, -3)$ ✓✓ Both coordinates.

$$\frac{d^2y}{dx^2} = 6x - 6$$

when $x = 0$ $\frac{d^2y}{dx^2} < 0 \therefore$ Max at $(0, 1)$ ✓ (Test)

when $x = 2$ $\frac{d^2y}{dx^2} > 0 \therefore$ Min at $(2, -3)$

(ii) let $\frac{d^2y}{dx^2} = 0$ to find pts of inflexion

$$6x - 6 = 0$$

$$x = 1$$

$\therefore$ at $(1, -1)$ ✓

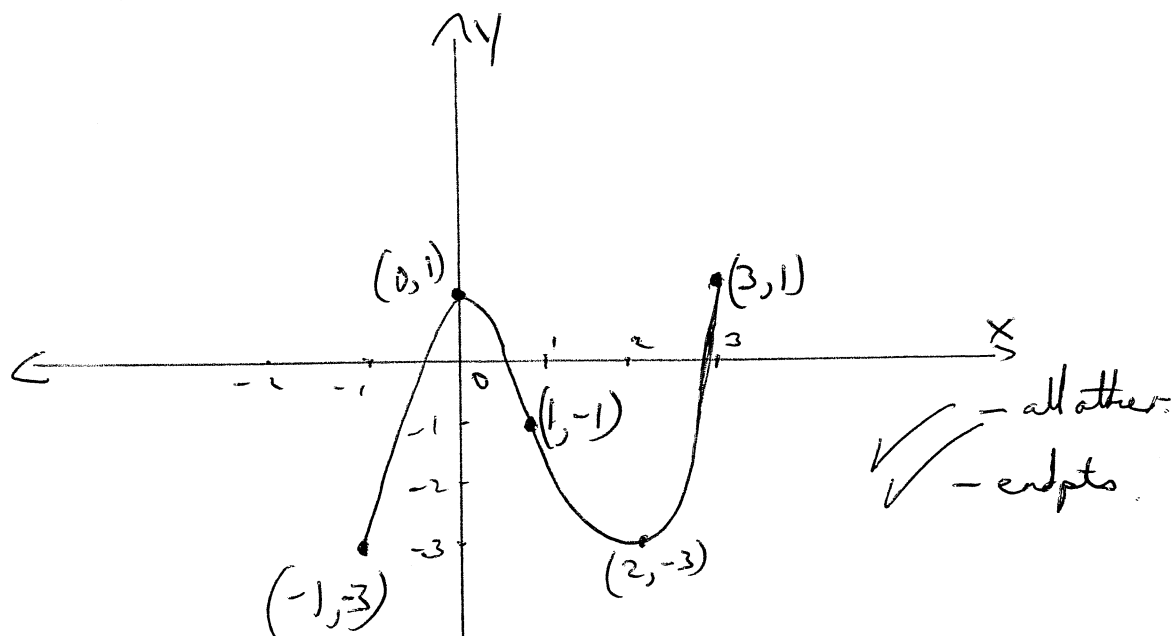
Creavity test

x	0.9	1	1.1
$\frac{d^2y}{dx^2}$	-	0	+

$\therefore$ change in concavity ✓

(iv) $x < 1$ concave down $[\frac{d^2y}{dx^2} < 0]$ ✓

(v)



✓✓ - all other
✓ - end pts.

(v) let $x^3 - 3x^2 + 1 = k$ ✓
 $\therefore -3 < k < 1$ ✓

Q4(a) $x^2 - 3x + 5 = A(x-1)^2 + B(x-1) + C$

$A = 1$ [coeff of x^2 is 1] ✓

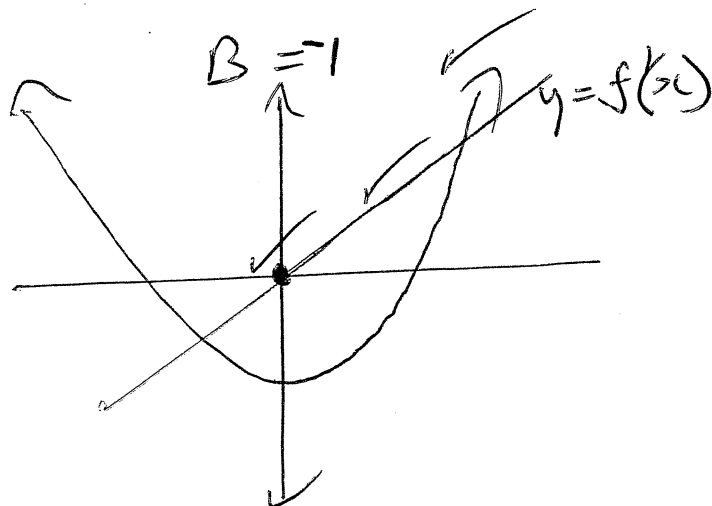
let $x=1 \therefore 1-3+5 = C$

$C = 3$ ✓

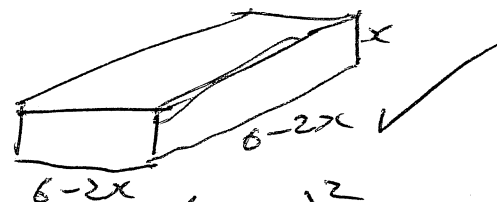
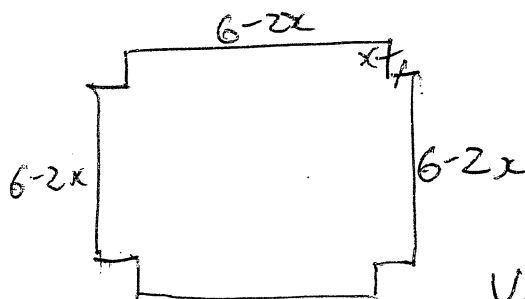
let $x=0 \therefore 5 = A + B + C$

$\therefore 5 = 1 + B + 3$

(b)



(c) (i)



$V(x) = x(6-2x)^2$ ✓

(ii) $V(x) = x(6-2x)^2$

$V'(x) = (6-2x)^2 - 4x(6-2x) = (6-2x)^2 - 24x + 8x^2$

$= (6-2x)[6-2x-4x]$ ✓

$= (6-2x)[6-6x]$ ✓

$\therefore x = 3, 1$ ✓

$\therefore$ Max when $x=1$

$\therefore$ squares 1m by 1m ✓

$V''(x) = -4(6-2x) - 24 + 16x$

$V''(1) = -4 \times 4 - 24 + 16 < 0$ $V''(3) = -4(6-6) - 24 + 48 > 0$ ✓